

Procedure of U-score Calculation for Bound-constrained Numerical Problems

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I. PROCEDURE FOR U-SCORE

For the U-score calculation, we follow the pairwise scoring procedure for UCOPs. Since UCOPs do not involve constraint violations, the scoring is based only on the objective-error trajectory. Each run stores the best-so-far objective error, denoted by Min_EV , after initialization and then at fixed evaluation intervals. As specified in the technical report, the objective error is recorded every $10D$ evaluations until MaxFEs . Since $D = 30$ and $\text{MaxFEs} = 10000D$, each run contains 1001 saved points, including the initial point. The procedure of U-score calculation is visually illustrated in Fig. 1.

Let $E_{a,r}^{(f)}(t)$ denote the saved Min_EV value of algorithm a , run r , function f , and sampling point t . The objective error is defined as

$$E_{a,r}^{(f)}(t) = f(\mathbf{x}_{a,r}^{\text{best}}(t)) - f^*, \quad (1)$$

where $\mathbf{x}_{a,r}^{\text{best}}(t)$ is the best-so-far solution at sampling point t , and f^* is the known optimum of the corresponding function. Following the numerical precision setting, objective-error values smaller than or equal to $\epsilon_{\text{EV}} = 10^{-8}$ are set to zero.

Let m be the number of compared algorithms and n the number of independent runs. For each function, all mn trials are compared pairwise. The number of unordered trial pairs is

$$N_{\text{pair}} = \frac{mn(mn - 1)}{2}. \quad (2)$$

For two trials x and y , let E_x^* and E_y^* denote their final saved objective errors at MaxFEs . Since a smaller objective error is better, the comparison function is defined as

$$\rho(u, v; \epsilon) = \begin{cases} 1, & u < v - \epsilon, \\ 0.5, & |u - v| \leq \epsilon, \\ 0, & u > v + \epsilon. \end{cases} \quad (3)$$

Here, $\rho(u, v; \epsilon)$ denotes the point awarded to the first trial when comparing u and v .

The pairwise accuracy point awarded to trial x against trial y is

$$A(x, y) = \rho(E_x^*, E_y^*; \epsilon_{\text{EV}}). \quad (4)$$

Thus, the trial with the smaller final objective error receives one accuracy point. If both final errors are equal within the tolerance ϵ_{EV} , both trials receive 0.5 points.

The accuracy score of algorithm a on function f is obtained by summing the pairwise accuracy points of its n trials against all other trials:

$$\mathcal{A}_a^{(f)} = \sum_{x \in \mathcal{T}_a^{(f)}} \sum_{\substack{y \in \mathcal{T}^{(f)} \\ y \neq x}} A(x, y), \quad (5)$$

where $\mathcal{T}_a^{(f)}$ is the set of trials of algorithm a on function f , and $\mathcal{T}^{(f)}$ is the set of all trials from all compared algorithms on function f .

The speed score is computed by comparing the time needed to reach a pairwise reference error level. For a pair of trials x and y , the reference level is defined as

$$\theta_E(x, y) = \max(E_x^*, E_y^*). \quad (6)$$

The first reaching index of trial x with respect to this reference level is then

$$\tau_E(x, y) = \min \{t \mid E_x(t) \leq \theta_E(x, y) + \epsilon_{\text{EV}}\}. \quad (7)$$

If the reference level is never reached, $\tau_E(x, y)$ is set to infinity.

The pairwise speed point awarded to trial x against trial y is

$$S(x, y) = \rho(\tau_E(x, y), \tau_E(y, x); 0). \quad (8)$$

Therefore, the trial that reaches the pairwise reference error level earlier receives one speed point. If both trials reach the reference level at the same sampling point, each receives 0.5 speed points.

The speed score of algorithm a on function f is computed as

$$\mathcal{S}_a^{(f)} = \sum_{x \in \mathcal{T}_a^{(f)}} \sum_{\substack{y \in \mathcal{T}^{(f)} \\ y \neq x}} S(x, y). \quad (9)$$

The U-score of algorithm a on function f is the sum of its accuracy and speed scores:

$$\mathcal{U}_a^{(f)} = \mathcal{A}_a^{(f)} + \mathcal{S}_a^{(f)}. \quad (10)$$

The final U-score over all $K = 29$ UCOP functions is

$$\mathcal{U}_a = \sum_{f=1}^K \mathcal{U}_a^{(f)}. \quad (11)$$

A larger U-score indicates better overall performance because it reflects both final objective-error quality and convergence

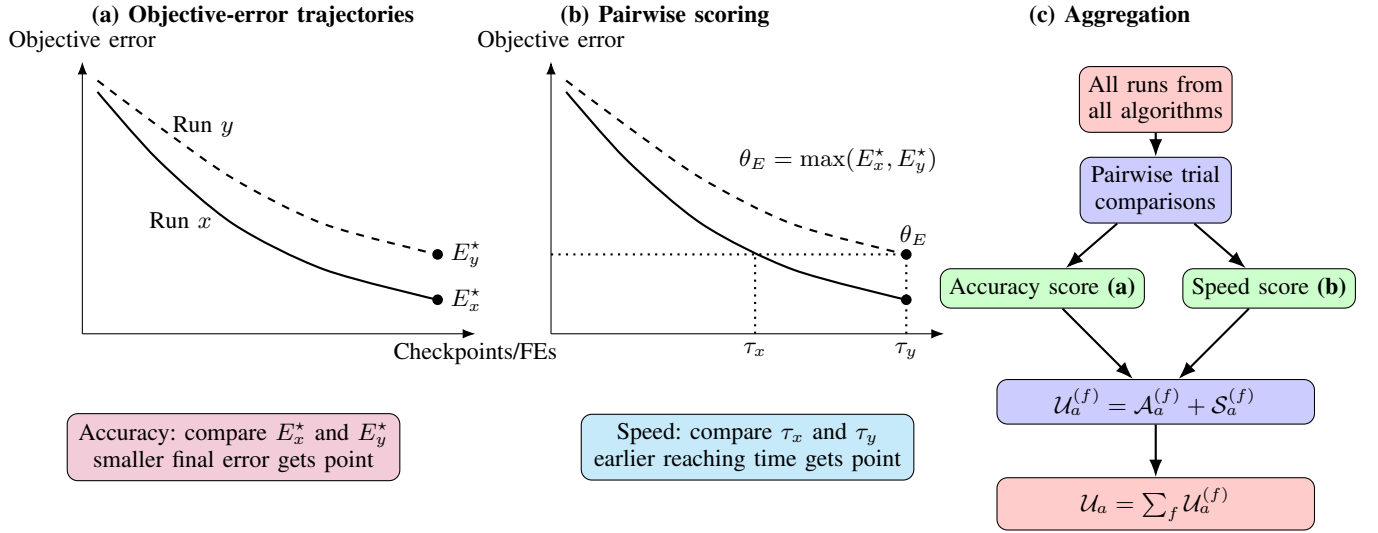


Fig. 1. Illustration of the pairwise U-score procedure for UCOPs. (a) Each run records the best-so-far objective-error trajectory at fixed checkpoints. (b) For a pair of runs, the accuracy component compares the final objective errors, while the speed component compares the first time each run reaches the pairwise reference error level $\theta_E = \max(E_x^*, E_y^*)$. (c) Pairwise accuracy and speed scores are accumulated over all trial pairs to obtain the function-wise U-score and then summed over all functions.

speed. In the reported results, the accuracy, speed, and U-score are obtained by summing $\mathcal{A}_a^{(f)}$, $\mathcal{S}_a^{(f)}$, and $\mathcal{U}_a^{(f)}$ over all test functions, respectively.